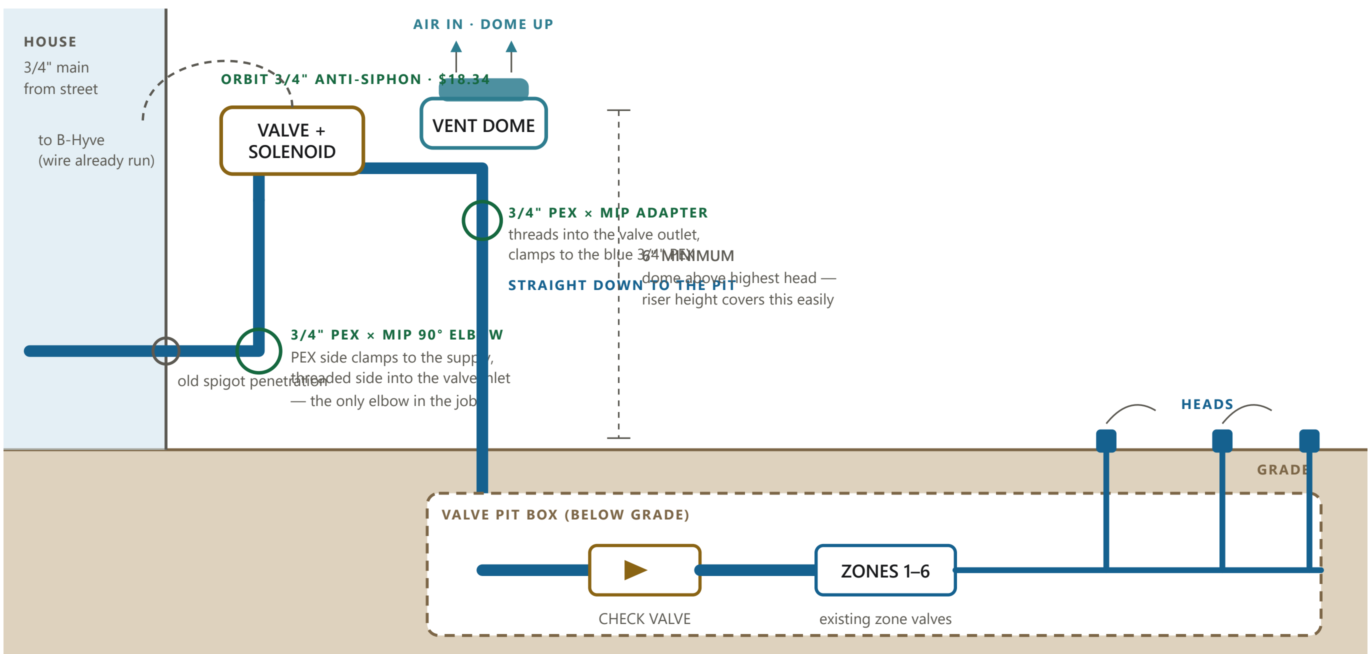
One valve, one elbow

Orbit 3/4" electric anti-siphon valve — master valve and backflow protection in a single body, $18.34. Replaces the plain Orbit already on the wall and reuses the B-Hyve wiring that's already run to it.



WHAT CHANGED FROM THE FIRST DRAWING

- **One part, not two.** The anti-siphon valve is the master valve and the backflow device in a single body. The earlier layout used your existing valve plus a separate breaker — two bodies, extra threads, extra fittings between them.
- **One elbow, not three.** The valve carries the flow up its inlet leg, across, and back down its outlet leg. Those turns happen inside the casting, so the only fitting you add is the 90 coming off the wall.
- **The vent is downstream of the valve seat by design.** Close the valve and the breaker is unpressurized — the continuous-pressure rule is satisfied by how the thing is built, not by arranging parts in a particular order.
- **Drop-in.** It replaces the plain Orbit already on that stub, and the B-Hyve wiring is already run to that exact spot for the existing solenoid.

PARTS

Orbit 3/4" electric anti-siphon valve	$18.34
3/4" PEX × 3/4" MIP 90° elbow — inlet side	have it
3/4" PEX × 3/4" MIP adapter — outlet side	have it
Stainless cinch clamps (matches existing joints)	have it
Check valve — already in the pit	have it
Zone valves, B-Hyve, wiring	have it

WATCH THESE FOUR

- **Dome up, above grade.** Non-negotiable — the poppet has to fall open on gravity.
- **Follow the cast arrows.** Inlet leg from the riser, outlet leg down to the PEX. Backwards, it won't protect anything.
- **Brace it.** The valve sits at the top of a riser and the PEX shouldn't carry its weight.
- **It's above ground.** Same freeze exposure as what's on that wall today, so treat it the same way at winterization.

THE ONE HONEST LIMIT

> Your six zone valves are still shutoff valves **downstream** of an atmospheric breaker, which the standard doesn't strictly allow. That's unchanged from every layout we've looked at — no better, no worse. The by-the-book alternative is a pressure vacuum breaker at roughly $80–150, plus annual testing by a licensed tester, which is exactly the attention you're trying not to attract. The **check valve in the pit** is a useful second layer but isn't a listed preventer on its own.

Revised 15 August 2026 · not to scale · supersedes the two-part layout drawn earlier the same day.